Start Here

how to read this

Every page uses the same colours. Learn them once and the rest of the document explains itself.

Colours come in two layers. One layer is us talking to you. The other is the maths colouring itself, so you can see an expression's parts at a glance.

what we tell you

orange	a new word, defined where it first appears
teal	the answer, or the idea that matters most
red	a mistake people make here. Read these twice.
white	ordinary explanation

inside the maths

blue	a letter standing for a number, such as x
green	a plain number
yellow	a function name, such as f or $\sin$

The two layers never share a colour. An orange word is always vocabulary. A blue letter is always a variable.

Step boxes carry the work. Each box is one step. The arrow below a box means "now do this next". Read them from top to bottom.

1. THIS IS A STEP BOX
The label says what the step does. The body shows the work.

↓

2. THE ARROW MEANS "NEXT"
Nothing is skipped between two boxes.

Use worked examples actively. Cover everything below the first box, try the step yourself, then uncover one box and compare. Reading a solved problem teaches far less than attempting it.

why algebra works

Four ideas carry all of algebra. Everything later is a consequence of them.

1. A LETTER IS A NUMBER YOU DO NOT KNOW YET
A **variable** is a name for an unknown number. In $x+3=7$, the letter x stands for one specific number. Algebra finds it.
(A variable is not a mystery. It is a number wearing a name tag.)

↓

2. AN EQUATION IS A CLAIM
The sign $=$ claims that both sides name the same number. $2+3=5$ claims that "$2+3$" and "5" are two names for one number.
The $=$ sign does not mean "here comes the answer". It means "these two are equal".

↓

3. THE BALANCE RULE
Picture a balance scale that is level. Add the same weight to both pans and it stays level. An equation behaves the same way.

Do the same thing to both sides, and the equation stays true.

This one rule licenses every step you will ever take when solving.

↓

4. A FUNCTION IS A MACHINE
A **function** takes a number in and gives exactly one number out. Write $f(x)=2x+1$. Put in 3, get out 7.
"Exactly one" is the whole rule. A machine that sometimes gives 7 and sometimes gives 9 for the same input is not a function.

Why solving works: you use the balance rule to peel operations off the variable, one at a time, until the variable stands alone. Nothing more mysterious than that is happening.

the rules everything assumes

Later topics use these without stopping to explain them. They are collected here once, in full.

Order of operations

When an expression mixes operations, everyone must read it the same way. The agreed order:

1. **Parentheses** — do anything inside grouping symbols first
2. **Exponents** — powers and roots
3. **Multiply and divide** — left to right, as they appear
4. **Add and subtract** — left to right, as they appear

Steps 3 and 4 are each one step, not two. Multiplication does not outrank division. In $12\div3\times2$, work left to right: $4\times2=8$. It is not $12\div6=2$.

WORKED: $2+3\times(8-6)^2$
parentheses: $8-6=2 \;\Rightarrow\; 2+3\times2^2$
exponent: $2^2=4 \;\Rightarrow\; 2+3\times4$
multiply: $3\times4=12 \;\Rightarrow\; 2+12$
add: 14

Signed numbers

adding, same sign	add, keep the sign: $-3+(-4)=-7$
adding, different signs	subtract, keep the bigger sign: $-7+3=-4$
subtracting	add the opposite: $5-(-3)=5+3=8$
multiply or divide	same signs $\to$ positive, different $\to$ negative

$-x$ does not mean "a negative number". It means "the opposite of x". If $x=-4$, then $-x=4$, which is positive.

Fractions

multiply	across the top, across the bottom: $\frac{2}{3}\cdot\frac{4}{5}=\frac{8}{15}$
divide	flip the second, then multiply: $\frac{2}{3}\div\frac{4}{5}=\frac{2}{3}\cdot\frac{5}{4}=\frac{10}{12}=\frac{5}{6}$
add or subtract	same bottom first: $\frac{1}{4}+\frac{1}{6}=\frac{3}{12}+\frac{2}{12}=\frac{5}{12}$

You may cancel factors, never terms. In $\frac{2x}{2}$ the 2 is a factor of the whole top, so it cancels. In $\frac{2x+3}{2}$ it is not, so it does not.

when you are stuck

Being stuck is almost always one of two things. Either you do not know what kind of problem this is, or you know but skipped a step. This process addresses both.

THE PROCESS
1. **Read the question again.** Write down what it asks for.
2. **Name the type.** Is this a linear equation, a factoring problem, a system?
3. **Pick the method** that matches that type.
4. **Work one step per line.** Never do two moves in one line.
5. **Check the answer.** Put it back into the original problem.

Step 5 is the one people skip, and it is the most valuable. Checking turns a guess into a certainty, and it catches arithmetic slips that no amount of re-reading will.

HOW TO CHECK — SUBSTITUTE AND COMPARE

Solve $3x-4=11$ and get $x=5$. Now put 5 back into the original equation:

$$3(5)-4=15-4=11 \quad \text{and the right side is } 11 \checkmark$$

Both sides give the same number, so $x=5$ is correct.

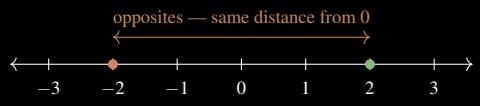
Check against the original problem, not your own second line. If you copied a sign wrongly in step one, checking against that line will happily confirm the mistake.

Still stuck after that? Work a simpler version of the same problem. Replace hard numbers with easy ones, solve that, then repeat your steps on the real numbers.

Pre-Algebra

integers and the number line

The **integers** are the whole numbers and their opposites: $\ldots, -2, -1, 0, 1, 2, \ldots$ Picture them on a line, with 0 in the middle and negatives to the left.



bigger further right. So $-2>-7$, because -2 sits to the right.

opposite same distance from 0, other side: the opposite of 5 is -5

absolute value distance from 0, so never negative: $|-7|=7$ and $|7|=7$

Negative numbers reverse your intuition about size. A debt of \$7 is a bigger debt than a debt of \$2, but -7 is the smaller number. Compare positions on the line, not the digits.

fractions

A **fraction** $\frac{a}{b}$ means "cut a whole into b equal parts, then take a of them". The top is the **numerator**, the bottom the **denominator**.

The one fact behind every fraction rule: multiplying the top and bottom by the same number does not change the value, because you are cutting into more pieces and taking proportionally more.

$$\frac{3}{4}=\frac{3\cdot2}{4\cdot2}=\frac{6}{8}=\frac{3\cdot5}{4\cdot5}=\frac{15}{20}$$

Simplifying

THE PROCESS

1. **Factor** the top and the bottom.
2. **Cancel** every factor they share.
3. **Multiply** what remains.

WORKED: SIMPLIFY $\frac{18}{24}$

$18=2\cdot3\cdot3$ and $24=2\cdot2\cdot2\cdot3$

shared factors: one 2 and one 3, so cancel 6 from each:

$$\frac{18}{24}=\frac{6\cdot3}{6\cdot4}=\frac{3}{4}$$

Mixed numbers

A **mixed number** like $2\frac{3}{5}$ means $2+\frac{3}{5}$. To calculate with it, turn it into a single fraction first.

WORKED: $2\frac{3}{5}$ TO A SINGLE FRACTION

the 2 wholes are $2\cdot5=10$ fifths, plus the 3 fifths already there:

$$2\tfrac{3}{5}=\frac{10+3}{5}=\frac{13}{5}$$

The four operations

multiply straight across: $\frac{2}{3}\cdot\frac{4}{5}=\frac{8}{15}$

divide flip the second, then multiply: $\frac{2}{3}\div\frac{4}{5}=\frac{2}{3}\cdot\frac{5}{4}=\frac{10}{12}=\frac{5}{6}$

add common denominator first

subtract common denominator first

Only adding and subtracting need a common denominator. Multiplying and dividing do not. This is the most common mix-up in this section.

1. FIND THE COMMON DENOMINATOR

$\frac{3}{4}-\frac{2}{5}$: the smallest number both 4 and 5 divide into is 20

↓

2. REWRITE BOTH FRACTIONS

$\frac{3}{4}=\frac{3\cdot5}{4\cdot5}=\frac{15}{20}$ $\quad$ $\frac{2}{5}=\frac{2\cdot4}{5\cdot4}=\frac{8}{20}$

↓

3. SUBTRACT THE TOPS, KEEP THE BOTTOM

$$\frac{15}{20}-\frac{8}{20}=\frac{15-8}{20}=\frac{7}{20}$$

The denominator does not change. You are counting twentieths, and 15 of them minus 8 of them is 7 of them.

The rules in general form

The worked examples above use numbers. Here are the same four rules with letters, which is what you apply when the fractions contain variables.

add $\frac{a}{b}+\frac{c}{d}=\frac{ad+bc}{bd}$

subtract $\frac{a}{b}-\frac{c}{d}=\frac{ad-bc}{bd}$

multiply $\frac{a}{b}\cdot\frac{c}{d}=\frac{ac}{bd}$

divide $\frac{a}{b}\div\frac{c}{d}=\frac{ad}{bc}$

cross multiply $\frac{a}{b}=\frac{c}{d}$ is the same as $ad=bc$

Why bd works as the bottom: multiplying the two denominators always produces a common denominator. It is not always the smallest one, so the answer may still need simplifying.

THE SAME SUM, BOTH WAYS

$\frac{1}{4}+\frac{1}{6}$ by the general rule: $\frac{(1)(6)+(1)(4)}{(4)(6)}=\frac{10}{24}=\frac{5}{12}$

by the smallest common denominator, 12: $\frac{3}{12}+\frac{2}{12}=\frac{5}{12}$

Same answer. The general rule needs no thinking and may need simplifying. The smallest denominator needs thinking and usually does not.

The add and subtract rules are the only ones with a $+$ or $-$ on top. Multiplying and dividing never build a sum, which is why they never need a common denominator.

decimals and percents

A **percent** means "per hundred". So 37% is $\frac{37}{100}$, which is 0.37. All three are the same number written three ways.

percent → decimal	move the point two left: $37\% \to 0.37$
decimal → percent	move the point two right: $0.05 \to 5\%$
decimal → fraction	write the place value, then simplify: $0.375 = \frac{375}{1000} = \frac{3}{8}$
fraction → decimal	divide top by bottom: $\frac{3}{8} = 3 \div 8 = 0.375$

The two percent questions

A **percent of a number** ⇒ turn the percent into a decimal and multiply

B **percent change** ⇒ $\frac{\text{new} - \text{old}}{\text{old}} \times 100\%$

WORKED (A): WHAT IS 15% OF 80?

$15\% = 0.15$, so $0.15 \times 80 = 12$

↓

WORKED (B): A PRICE RISES FROM 40 TO 50

$$\frac{50-40}{40} \times 100\% = \frac{10}{40} \times 100\% = 0.25 \times 100\% = 25\% \text{ increase}$$

Always divide by the *old* value. Going $40 \to 50$ is a 25% rise, but going $50 \to 40$ is a 20% fall. The same 10 difference gives two different percents, because the starting points differ.

ratios and proportions

A **ratio** compares two amounts, written $3 : 4$ or $\frac{3}{4}$. A **proportion** sets two ratios equal.

THE PROCESS — SOLVE A PROPORTION

1. **Write** the proportion, keeping matching units in matching positions.
2. **Cross multiply** — top left times bottom right, and the other pair.
3. **Solve** the resulting simple equation.

WORKED: $\frac{3}{4} = \frac{x}{20}$

cross multiply: $3 \cdot 20 = 4 \cdot x \quad \Rightarrow \quad 60 = 4x$

divide both sides by 4: $x = 15$

check: $\frac{15}{20} = \frac{3}{4}$ ✓

Why cross multiplication is legal: it is the balance rule twice. Multiply both sides by 4, then by 20. Cross multiplying just does both at once.

A **unit rate** is a ratio with 1 on the bottom. Divide to find it: 150 miles in 3 hours is $\frac{150}{3} = 50$ miles per hour.

exponents and square roots

An **exponent** counts repeated multiplication: $2^5 = 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 = 32$. Read 2^5 as "two to the fifth".

An exponent is not multiplication. 2^5 is 32, not 10. The exponent says how many times to multiply, not what to multiply by.

A **square root** asks the reverse question: $\sqrt{49} = 7$ because $7^2 = 49$. Memorize these, because they appear constantly:

$1^2 = 1 \quad 2^2 = 4 \quad 3^2 = 9 \quad 4^2 = 16 \quad 5^2 = 25 \quad 6^2 = 36$
$7^2 = 49 \quad 8^2 = 64 \quad 9^2 = 81 \quad 10^2 = 100 \quad 11^2 = 121 \quad 12^2 = 144$

Estimating a root: trap it between two perfect squares. $\sqrt{50}$ sits between $\sqrt{49} = 7$ and $\sqrt{64} = 8$, and much nearer 7, so about 7.1.

Scientific notation

Scientific notation writes any number as one digit, a decimal point, then a power of ten. It keeps very large and very small numbers readable.

$$4{,}500{,}000 = 4.5 \times 10^6 \qquad 0.00032 = 3.2 \times 10^{-4}$$

THE RULE

1. **Move the point** until exactly one non-zero digit sits in front of it.
2. **Count the moves.** That count is the exponent.
3. **Moved left** gives a positive exponent. **Moved right** gives a negative one.

A negative exponent does not mean a negative number. 3.2×10^{-4} is small and positive. The sign describes the size, not the direction from zero.

To multiply, handle the two parts separately: $(2 \times 10^3)(4 \times 10^5) = (2 \cdot 4) \times 10^{3+5} = 8 \times 10^8$

basic geometry

perimeter	distance around the outside — add the sides
area	flat space covered — measured in square units
volume	space filled — measured in cubic units

shape	area	perimeter
rectangle	$A = lw$	$P = 2l + 2w$
triangle	$A = \frac{1}{2}bh$	add the three sides
circle	$A = \pi r^2$	$C = 2\pi r$

Box volume: $V = lwh$. A box 4 by 3 by 2 holds $4 \cdot 3 \cdot 2 = 24$ cubic units.

A circle has a circumference, not a perimeter, and πr^2 is the area while $2\pi r$ is the distance around. Mixing these two is the classic circle mistake.

The Pythagorean theorem

In a right triangle only, the two short sides (the **legs**) and the longest side (the **hypotenuse**) satisfy

$$a^2 + b^2 = c^2 \qquad \text{where } c \text{ is always the hypotenuse}$$

WORKED: LEGS 3 AND 4, FIND THE HYPOTENUSE

$3^2 + 4^2 = 9 + 16 = 25$, so $c = \sqrt{25} = 5$

↓

WORKED (BACKWARD): HYPOTENUSE 13, ONE LEG 5

the missing side is a leg, so it belongs on the left:

$$5^2 + b^2 = 13^2 \;\Rightarrow\; 25 + b^2 = 169 \;\Rightarrow\; b^2 = 144 \;\Rightarrow\; b = 12$$

Put the hypotenuse alone on the right. If you place a leg where c belongs, you will get a larger answer than the hypotenuse, which is impossible.

Legend

Colours: what we tell you
orange — a new word
teal — the answer
red — a common mistake
white — explanation
boxes stack top to bottom
↓ means "next step"
Colours: inside the maths
blue — a letter x, y
green — a number $3, 7$
yellow — a function f, g
pink — brackets ( )
grey — signs $+ - =$
Comparing
$=$ is equal to
$\neq$ is not equal to
$<$ is less than
$>$ is greater than
$\leq$ less than or equal
$\geq$ greater than or equal
$\approx$ is about
Operations
$\cdot$ times
$\pm$ plus or minus
$\sqrt{\ }$ square root
$\sqrt[n]{\ }$ nth root
$|x|$ absolute value
x^{-1} one over x
f^{-1} inverse function
Names
π about 3.14159
θ an angle
i $\sqrt{-1}$
m slope
a_n the nth term
$\Rightarrow$ therefore